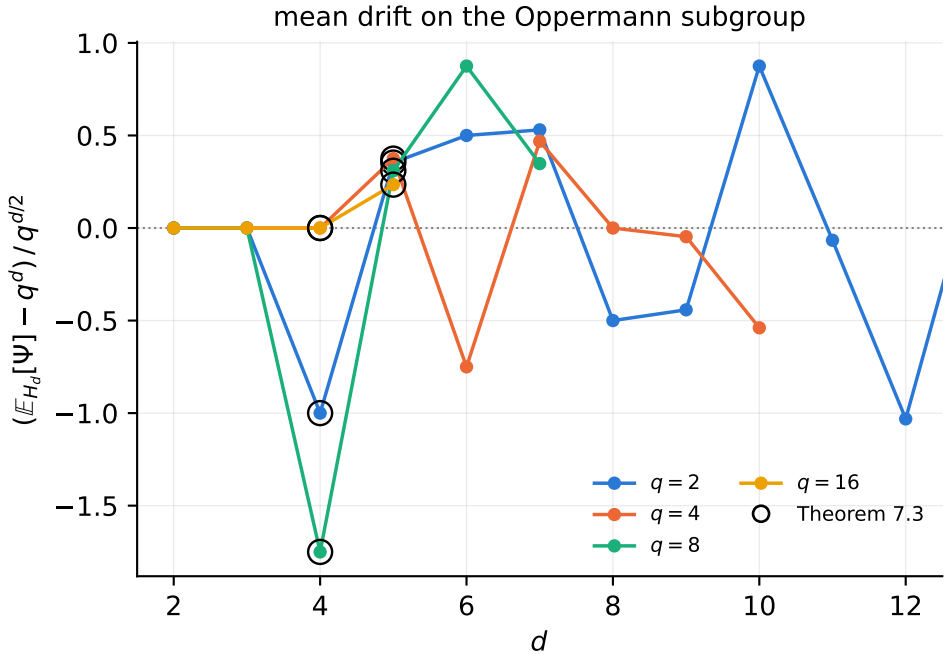
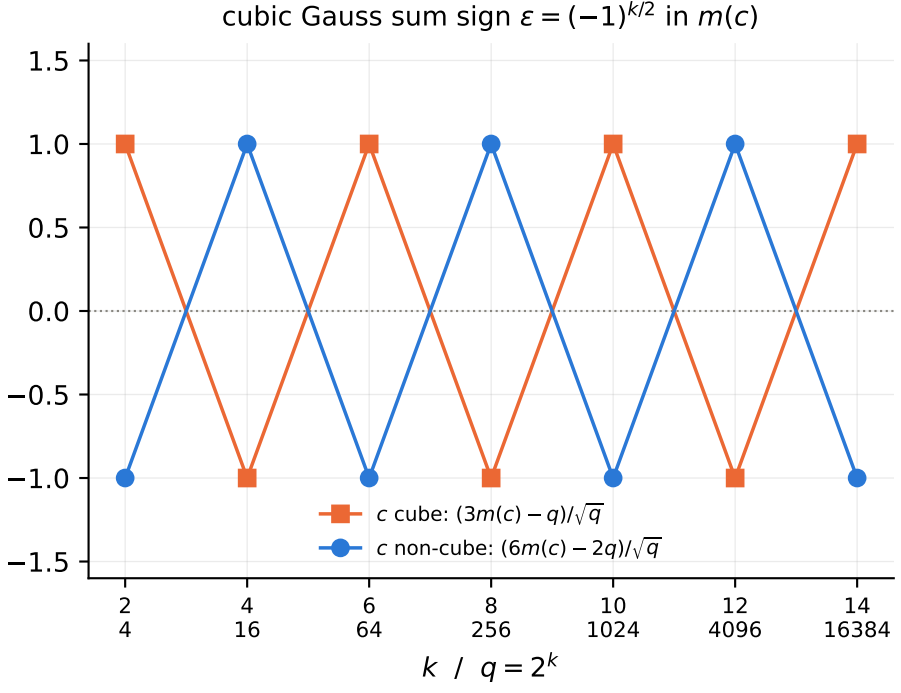

mean drift on the Oppermann subgroup
$(\mathbb{E}_{H_d}[\Psi] - q^d)\,/\,q^{d/2}$
1.0
0.5
0.0
−0.5
−1.0
−1.5
2
4
6
8
10
12
d
$q = 2$
$q = 4$
$q = 8$
$q = 16$
Theorem 7.3


cubic Gauss sum sign $\varepsilon = (-1)^{k/2}$ in $m(c)$
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
2 4
4 16
6 64
8 256
10 1024
12 4096
14 16384
k / $q = 2^k$
c cube: $(3m(c) - q)/\sqrt{q}$
c non-cube: $(6m(c) - 2q)/\sqrt{q}$